

Rodrigo Nahuel Castillo AI

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PROFESSIONAL SUMMARY

AI Agent Developer specialized in multi-agent architecture, prompt engineering, and tool integration via MCP on top of LLMs. Led the migration of a large-scale government conversational assistant to an agent architecture, with custom tooling in TypeScript and Python. I apply QA Automation practices to agent validation: I build test sets, measure intent coverage and collisions, and validate model behavior before every production release.

EXPERIENCE

Semi-Senior AI Agent Developer

Botmaker

April 2025 – Present, Buenos Aires, Argentina

- Led the migration of a government conversational assistant from a static flow architecture to an agent architecture, completing the structural analysis and the full migration plan in 1 month.
- Reorganized 9,288 scattered intents across 47 government areas into an ecosystem of 37 domain-specialized agents plus 4 cross-cutting groups, replacing rigid decision trees with agentic orchestration.
- Audited the bot's complete inventory of 2,802 variables: detected 745 ghost variables and flagged 232 for removal; the analysis was escalated as a formal report to the agency's authorities and enabled the cleanup of accumulated technical debt.
- Took the orchestrator and six specialized agents of the ecosystem to production (org chart, subway, driver's licenses, tourism, onboarding, and fallback) on the assistant's WhatsApp and web channels, sustaining its iteration with the agency once live.
- Designed the prompt architecture of the orchestrator and the domain-specialized agents, eliminating rule contradictions that caused formatting violations in production, and implemented accessibility fixes to the live prompt for correct use with screen readers.
- Validated agent behavior before the production release with custom Python test sets: designed the triggers of 18 new flows against a corpus of 526 real phrases, measuring recall per intent and confirming zero collisions between intents; also documented the model selection for a document analysis use case in a formal validation report comparing two LLMs.
- Developed custom MCP tools in TypeScript: semantic search of related topics with TF-IDF and cosine similarity over 676 indexed topics from 37 agents, and hierarchical queries over structured data, expanding the agents' reach without adding manual flows.
- Standardized the tone of the entire conversational ecosystem with a style guide derived from the analysis of more than 600 use case documents, and automated knowledge base construction with Python pipelines, replacing manual document-by-document assembly.
- Automated internal HR team processes (pre-onboarding, leave, and internal requests) with WhatsApp Flows and integrations with messaging and corporate spreadsheets, eliminating manual form handling.
- Detected critical security and performance failures in an enterprise client's production tool and proposed its redesign into a deterministic solution.

- Packaged the prompt creation process into reusable skills, now adopted by the team for other clients' agents.
- Implemented a monitoring and early-warning system for agent performance, using latency and error rate metrics to predict and mitigate operational issues before they impact users.

Junior QA Tester / Testing Analyst

QA Actions Group

September 2024 – March 2025, Buenos Aires, Argentina

- Automated regression suites with Tricentis Tosca on critical systems, replacing repetitive manual execution and shortening the validation cycle of each release.
- Detected and documented defects at different stages of the SDLC, following up until resolution with development teams under Agile and Scrum.
- Proposed the re-engineering of a critical tool for an enterprise client, identifying security and performance flaws and suggesting a deterministic solution to improve its stability.

EDUCATION

Computer Science Engineering

National University of La Matanza (UNLAM) · 2022 – Present

- 3rd year.

Electronics Technician

Technical Secondary School No. 6 · 2015 – 2021

CERTIFICATIONS

Tricentis Tosca Automation Specialist Level 1 (AS1)

Tricentis · 2024

Tricentis Tosca Automation Specialist Level 2 (AS2)

Tricentis · 2024

SKILLS

AI & Agents: multi-agent architecture, agentic workflows, prompt engineering, MCP (Model Context Protocol), RAG, knowledge bases, state management, intent and entity design, Botmaker, n8n

LLMs: Anthropic (Claude), OpenAI (GPT), Google Gemini; API consumption and translation of business rules into model behavior

Languages: Python, TypeScript, JavaScript / Node.js, SQL

Backend and Integrations: REST APIs, webhooks, client-server architecture, WhatsApp Flows, Google Apps Script, automation scripting

Data and Cloud: Relational SQL, MongoDB, data modeling, Google Cloud Platform, Google Workspace, Git / GitHub / GitLab
